# PROJECT TECHNICAL REPORT

DRAFT

| GENERAL INFORMATION | | |
|---|---|---|
| CLIENT | Kimfor | |
| TECHNICAL SUPPORT | Document developed by QualityNet | qualitynet® Esperti in Certificazioni |
| | In collaboration with EcamRicert Srl \| Mérieux NutriSciences | Mérieux NutriSciences |
| TECHNICAL AND REGULATORY REFERENCES | UNI EN ISO 14040:2021<br>UNI EN ISO 14044:2021<br>PCR 2019:14 Construction products (EN 15804+A2) (2.0.1) | |
| VERSION | 01 | |
| DATE | 10/02/2026 | |

DRAFT

INTRODUCTION ..... 2
COMPANY ..... 3
CONTEXT ..... 4
LIFE CYCLE APPROACH ..... 5
1. OBJECTIVE OF THE STUDY ..... 6
2. SCOPE OF THE STUDY ..... 8
2.1. LIFE CYCLE STAGES AND INFORMATION MODULES ..... 8
2.2. DECLARED UNIT ..... 8
2.3. REFERENCE SERVICE LIFE (RSL) ..... 8
2.4. SYSTEM BOUNDARIES ..... 8
2.4.1. TEMPORAL, GEOGRAPHICAL, AND TECHNOLOGICAL BOUNDARIES ..... 9
2.4.2. ORGANIZATIONAL BOUNDARIES ..... 10
2.4.3. OPERATIONAL BOUNDARIES ..... 10
2.5. EXCLUSION CRITERIA AND ASSUMPTIONS ..... 10
2.6. DATA SELECTION ..... 11
2.7. DATA QUALITY REQUIREMENTS ..... 11
3. LIFE CYCLE INVENTORY (LCI) ..... 13
3.1. DATA COLLECTION ..... 13
3.2. CALCULATION PROCEDURES ..... 13
3.3. REFINEMENT OF SYSTEM BOUNDARIES ..... 14
3.4. ALLOCATION OF INPUT FLOWS AND OUTPUT EMISSIONS ..... 14
4. LIFE CYCLE IMPACT ASSESSMENT (LCIA) ..... 15
4.1. SELECTION OF IMPACT CATEGORIES, CATEGORY INDICATORS, AND CHARACTERIZATION MODEL ..... 15
4.2. CLASSIFICATION ..... 16
4.3. CHARACTERIZATION ..... 16
5. LCA RESULTS ..... 18
5.1. PRESENTATION OF RESULTS ..... 18
5.1.1. THERMOLOCK THERMAL INSULATION PLASTER ..... 18
6. BIOGENIC CARBON CONTENT IN THE PRODUCT AND ITS PACKAGING ..... 20
7. RESULTS ANALYSIS AND AVERAGE PERCENTAGE CONTRIBUTIONS ..... 21
8. CONCLUSIONS, LIMITATIONS, AND RECOMMENDATIONS ..... 23
BIBLIOGRAPHY ..... 24
WEB REFERENCES ..... 24
*ANNEX A: DATA AND DATASET QUALITY* ..... 25

DRAFT

# INTRODUCTION

This study was conducted for Kimfor and was carried out by QualityNet (in collaboration with EcamRicert Srl).
This report presents the Life Cycle Assessment (LCA) study performed on one product: Thermolock Thermal Insulation Plaster.

The analysis was conducted following the main guidelines of standards UNI EN ISO 14040:2021, UNI EN ISO 14044:2021, and UNI EN 15804:2019. For the calculation procedures required by these standards, the software SimaPro v10.3.0.1 was used. It should be noted that, to ensure reliability and traceability of information as well as to certify compliance with technical standards, it is required to entrust the verification of any environmental declaration regarding LCA studies to an independent third-party entity. For this reason, this report should not be used by sponsors or readers to make specific claims about systems or product schemes unless such claims are clearly supported by the Life Cycle Inventory (LCI) results and are accompanied by a reference to the full publicly available report. The use of the study results for advertising purposes must comply with and be consistent with the principles indicated in the ISO 14040 series guidelines.

QualityNet, *EcamRicert Srl, and their members are not responsible for any use of the study results by third parties in a manner that is not fully compliant with the guidelines described herein.*

DRAFT

## COMPANY

KIMFOR IZOLASYON has been operating since 2009 in Bursa, the industrial capital. It is a leading company in the development, production, and introduction of new generation ecological insulation systems to the international market.

Kimfor's product portfolio mainly includes insulation coatings, protective layers, and chemical formulations designed for use in residential, commercial, and industrial buildings. These products are intended to improve building performance by contributing to thermal resistance, moisture control, fire protection, and noise reduction, depending on the specific application and formulation.

DRAFT

# CONTEXT

The globalization of markets, digital connectivity, and the scarcity of natural resources contribute to increasing the complexity of the environment in which companies operate (Sebastiani, 2013). The uncertainties arising from this complexity make it increasingly difficult to predict rapid market changes; therefore, companies are progressively expressing the need to develop both their resilience and their flexibility to face this unpredictable future (Sebastiani, 2013).

To succeed, or even simply to continue operating regularly, companies must adopt an increasingly strategic approach capable of supporting, from a long-term perspective, the sudden changes that may occur (Sebastiani, 2013). It is in this direction that the concept of corporate sustainability is emerging: a business approach capable of integrating sustainability strategies and practices into traditional business mechanisms, with the aim of creating value not only from an economic point of view but also in social and environmental terms (Figure 1). It is a "business philosophy that deeply and transversally cuts across the entire organizational structure and, in many cases, requires a radical transformation of business models, starting from a redefinition of the company's DNA and leading to a thorough green revision (energy and water consumption, emissions, waste, resource use, efficiency improvements, etc.) of processes and products" (M. Fasan, 2017).

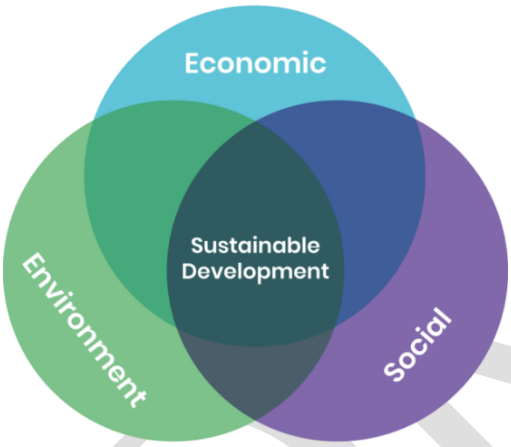



*Figure 1: Sustainability as a balance between economic costs, environmental impacts, and collective benefits*

This approach enables companies to explore new ways to compete in an increasingly complex reality. A sustainable company is therefore a well-structured and organized entity that, from a long-term perspective, aims to create a balance among the economic, social, and environmental dimensions (Sebastiani, 2013).

According to the "Three P's" model: Planet, People, Profit (Elkington, 1997), sustainable development at the company level is achieved when profits are compatible with environmental integrity and social equity. The Planet dimension refers to the efficient management of natural resources and the optimization of the environmental impacts associated with products from a life cycle perspective (M. Fasan, 2017). The People dimension aims to ensure opportunities and resources for all members of society, with particular attention to the health and safety of workers and consumers.

Finally, the Profit dimension focuses on ensuring economic prosperity while maintaining an adequate competitive position (Siano, 2012).

In this sense, companies that adopt a sustainable business approach gain more competitive positions: the integration of the three sustainability dimensions within business processes leads companies to maintain constant attention to the social and environmental context in which they operate, to build relationships with an increasing number of stakeholders, and thus to be more responsive (and flexible) to continuous changes.

DRAFT

## LIFE CYCLE APPROACH

The Life Cycle Approach (LCA) makes it possible to map, understand, and reduce the potential environmental risks arising from the manufacturing of products and from the activities of organizations and services. The relevance of this method lies in analyzing the phases of a production process, an organization, or a service as interconnected and correlated, becoming a fundamental design support tool for development and improvement in relation to:

A. **Improve**: qualitative selection of the parameters to be measured.

B. **Catalogue**: quantitative evaluation of the parameters related to unit processes within the system boundaries, in relation to the reference unit. The result is a structured design matrix organized by environmental performance indicators.

C. **Analyze**: assessment of the environmental effects connected to environmental performance indicators to identify potential environmental risks of the examined system. This is followed by an analysis of the obtained results to act on the study setup and define possible intervention strategies to improve its control.

The described methodological approach is foreseen in environmental management standards, which represent the design foundation for specific guidelines and major sustainability protocols. At an international level, the LCA methodology is gaining strategic importance as a scientifically suitable and recognized tool for framing significant environmental aspects through voluntary environmental policy instruments.

DRAFT

# 1. OBJECTIVE OF THE STUDY

The presented study, of the type "from cradle to gate with modules C1–C4 and module D," considered the information and data characterizing the mandatory information modules (see section 5, UNI EN 15804:2019).

The objective of the analysis is to calculate the eco-profile attributable to the manufacturing phase of the products. To achieve this goal, Kimfor intends to start from the LCA analysis attributable to the production of one unit of finished product, including its packaging.

The company plans to conduct the analysis on one product types:

- Thermolock Thermal Insulation Plaster

  Product description: Low λ (0.060 W/mK), breathable, waterproof, A1 fireproof, F360; low emissivity and high solar reflectivity.

| Product composition in terms of mass (Kg) | | | % | |
|---|---|---|---|---|
| Material: | Cement | 0,2 | 20% | PRODUCT |
| Material: | Mineral Filler (calsite/quartz/perlite) | 0,75 | 75% | |
| Material: | Polymer | 0,03 | 3% | |
| Material: | Additives (hydrophobic, etc.) | 0,02 | 2% | |

| Product composition in terms of mass (Kg) | | | % | |
|---|---|---|---|---|
| Material: | Kraft | 0,0064 | | PACKAGING |
| Material: | Wooden | 0,027 | | |
| Material: | Nylon | 0,00093 | | |

*Figure 2: Product Image*

DRAFT

The company aims to optimize the design of its products from an environmental sustainability perspective, enabling it to guide market choices by investing in systems with lower resource consumption, while controlling and mitigating both direct and indirect environmental impacts in line with an integrated policy approach.

The chosen analytical approach is top-down, starting from the definition of the general system boundaries and then moving into the details of each unit process included within those boundaries.

The target audience of the study comprises stakeholders, in particular other companies operating in the same sector. The study is therefore B2B (Business to Business).

The main objective of this Life Cycle Assessment (LCA) study is to support the company in developing informed environmental strategies, while also providing stakeholders with transparent and detailed information on the environmental impacts associated with the analyzed products. An additional purpose of the study is the preparation of an Environmental Product Declaration (EPD) in compliance with the requirements of the International EPD® System Program Operator, as established by the General Programme Instructions and the relevant Product Category Rules (PCR). Furthermore, the LCA study was conducted to meet the requirements set out by the Minimum Environmental Criteria (CAM), thus contributing to demonstrating the environmental compliance of the product.

Obtaining an EPD represents not only a technical tool for environmental reporting and regulatory compliance but also a strategic lever for competitive positioning. The decision to pursue EPD certification is therefore part of a broader strategy aimed at enhancing the environmental value of products and strengthening the company's brand image, creating new business opportunities in a context of growing focus on sustainability.

DRAFT

# 2. SCOPE OF THE STUDY

## 2.1. LIFE CYCLE STAGES AND INFORMATION MODULES

In accordance with the requirements of the reference documents and the followed PCR (PCR 2019:14 CONSTRUCTION PRODUCTS v2.0.1), the adopted approach and the environmental information provided have been divided by information modules.

## 2.2. DECLARED UNIT

For the purpose of calculating the eco-profile covered by this study, the products manufactured by Kimfor were analyzed. The declared unit serves as the reference for the mathematical normalization of material and energy flows included in the investigated information modules, in order to produce data and information expressed on a common basis. The declared unit thus constitutes the reference for combining the flows attributed to the object of the analysis and for combining the environmental impacts related to the specified modules.
In accordance with the directives of the reference standard and the product rule, the declared unit is defined as: 1 kg.

## 2.3. REFERENCE SERVICE LIFE (RSL)

With reference to what is indicated in the reference PCR, the reference service life (RSL) has not been addressed. As stated in the PCR: 'Note that the RSL of a product category is the reference period to which the performance of a product may be related in the functional unit (for a definition and description of RSL, see Section A.2.1 of the GPI and Section 6.3.4.1 of EN 15804).

## 2.4. SYSTEM BOUNDARIES

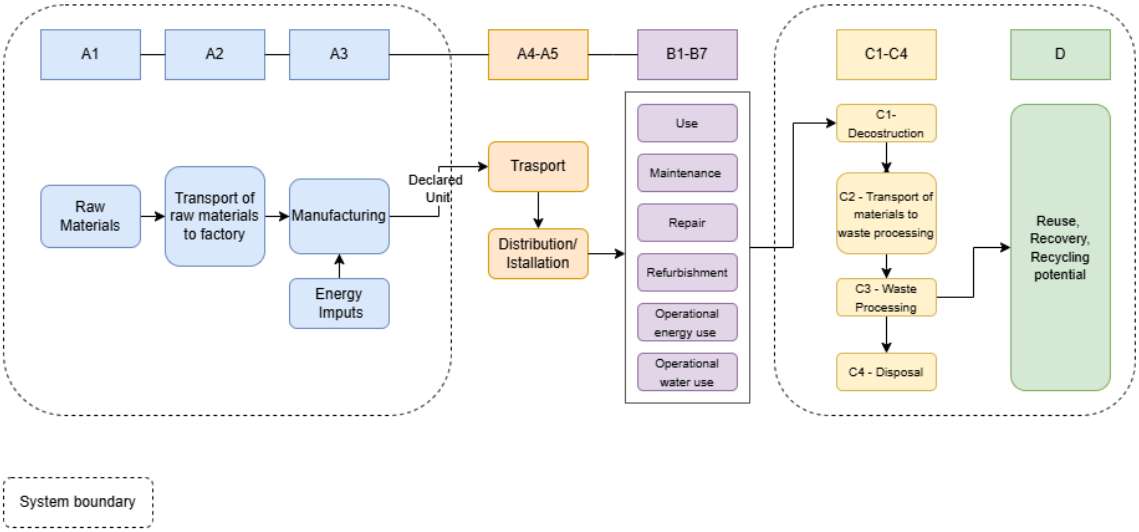
The purpose of defining the system boundaries is to establish a spatial, temporal, and operational scope within which to collect reliable data that reflect the actual environmental performance of the system and provide a complete description of it.
The detail and extent of the study are determined by these boundaries, which allow the identification of the unit processes to be included in the model. Along these unit processes, the main consumption factors involved in the production of the products were identified, and the calculation model was built and analyzed accordingly.

DRAFT

*Table 1: System boundaries*

| | PRODUCT STAGE | | | CONSTRUCTI ON PROCESS STAGE | | USE STAGE | | | | | | | END OF LIFE STAGE | | | | BENEFITS AND LOADS BEYOND THE SYSTEM BOUNDARIES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | y material supply | Transport | Manufacturing | Transport | Construction installation | Use | Maintenance | Repair | Replacement | Refurbishment | Operational energy use | Operational water use | Deconstruction, demolition | Transport | Waste processing | Disposal | Reuse-recovery-recycling potential |
| | A1 | A2 | A3 | A4 | A5 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | C1 | C2 | C3 | C4 | D |
| | X | X | X | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND | ND |
| Geography | TR | TR | TR | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Specific data used | 16,66% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Variation – product | 0% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Variation – site | 0% | | | - | - | - | - | - | - | - | - | - | - | - | - | - | - |

*Figure 3: System boundaries*




### 2.4.1. TEMPORAL, GEOGRAPHICAL, AND TECHNOLOGICAL BOUNDARIES

The temporal boundaries cover the period from 1st January 2024 to 31rd December 2024, a timeframe considered representative of the company's activities, and the study results should be interpreted within these boundaries. This period was chosen due to the most complete availability of information. relevant to the study.

The geographical boundaries of the study are defined Turkish territory (particularly for the manufacturing phase).

The technological boundaries refer to the average technological level specific to the temporal and geographical context described above.

### 2.4.2. ORGANIZATIONAL BOUNDARIES

In defining the organizational boundaries, the first step was to identify the organization of Kimfor and the facilities that make up the company, with regards to the unit processes relevant to the products under study. This means defining the organizational boundaries, namely the responsibilities attributable to Kimfor concerning the life cycle analysis, according to the boundaries already described.

The approach followed is the control approach, since the purpose of the study is to monitor and improve the environmental performance of the analyzed products. The analytical approach made it possible to study the products by evaluating them as systems affected by input and output factors, and consequently subject to work cycles that cause environmental impacts, in accordance with the reference PCR.

It should be noted that Kimfor activities take place at the facility located at:

- Yunuseli Mh. Akyıldız Cd. No: 3 Osmangazi / Bursa / Turkey

### 2.4.3. OPERATIONAL BOUNDARIES

The unit processes related to the production facility have been detailed. Within these boundaries, the material and energy flows characteristic of the unit processes, which can be categorized according to environmental performance indicators, were considered, and environmental impacts were subsequently calculated on these flows.

Characteristic data that could imply an environmental impact were therefore collected. These data, for the information module "[A3] Manufacturing", are mostly site-specific. This is an important feature, as it increases the reliability of the study and makes the calculation model a more realistic representation of the production process under analysis.

The operational processes can be summarized as follows:

*Figure 4: Production process*

## 2.5. EXCLUSION CRITERIA AND ASSUMPTIONS

The main assumptions are listed below:

- C1: Deconstruction / Demolition

The environmental impact is assumed to be very small and can be neglected since the de-construction and/or dismantling of insulation products take part of the demolition of the entire building

- C2: Transport to Disposal Site

This stage includes the transportation of the discarded construction products to final disposal. Average distance from demolition site to final disposal site is assumed to be 80 km as suggested by the used PCR (2019:14 v.2.0.1)

- C3: Waste Processing

Demolition waste is assumed not to be further processed and sent to landfill for final disposal.

- C4: Disposal

The insulation plaster is part of demolition waste and assumed to be 100% landfilled.

- D: Benefits and Loads

There potential benefit and loads have been estimated according to the fact that the products go completely to the landfill at the end of life. Benefit from packaging recycling is not taken into account, since there is not evidence of the final treatment of the packaging.

## 2.6. DATA SELECTION

In selecting the data to be used for the study, an effort was made to prioritize primary data that could be cataloged by the company. These data constitute the primary source of information for the inventory analysis. They are grouped according to environmental performance indicators, to which the results of the environmental performance will subsequently be referred. The identifiable environmental performance indicators are shown in Table 2.

*Table 2: Environmental performance indicators divided by information modules*

| Module | Description | Indicator | |
|---|---|---|---|
| A1 | Raw material supply | Raw material<br>Electricity consumption | UPSTREAM |
| A2 | Transport | Transport of raw materials and packaging | |
| A3 | Manufacturing | Material (packagings)<br>Water consumption<br>Transport of generated waste<br>Treatment of generated waste | CORE |
| C1 | De-construction demolition | Consumption linked to demolition | END OF LIFE |
| C2 | Transport | Waste transport | |
| C3 | Waste processing | Waste treatment | |
| C4 | Disposal | Disposal | |
| D | Benefits and loads beyond the system boundary | Potential for reuse, recovery, and recycling<br>Energy recovered from end-of-life<br>Secondary materials returned to the system | OUTPUT BEYOND SYSTEM BOUNDARY |

## 2.7. DATA QUALITY REQUIREMENTS

The choice of the indicators listed above was based on the fact that they are characteristic of the system examined within the described boundaries. Moreover, with particular attention to Kimfor's production processes, these indicators are:

- physically measurable and traceable by the company through appropriate documents;
- monitorable and potentially controllable by the company;
- represent an economic cost for the company.

The characteristics of the mass and energy flow data for the environmental performance indicators listed above, where technically possible, reflect compliance with specific quality requirements, such as:

DRAFT

- relation to a specific time period (time-related coverage);
- relation to a specific geographical context (geographical coverage);
- relation to a certain technological level (technology coverage);
- defined degree of precision, completeness, representativeness, consistency, and reproducibility;
- characterized by a data source (sources of the data);
- defined degree of uncertainty (uncertainty of the information).

The quality requirements of the collected data are shown in Table 3.

*Table 3: Data quality requirements according to UNI EN 15804*

| QUALITY REQUIREMENTS | COMPLIANCE |
|---|---|
| Temporal Coverage | The data were collected during the year to which they refer and cover its entire duration (2024). At the time of the study, these data are generally the most recent and complete available. |
| Geographical Coverage | The data were collected in the geographical area to which they refer, in accordance with the Geographical Coverage defined requirements. Regarding geographical coverage, preference was given to using data in the following order of priority: local, national, European. |
| Technical Coverage | Average technological level in line with the defined temporal and geographical boundaries. |
| Completeness | All significant material and energy flows have been analyzed and accounted for. The data relating to raw materials are expressed in kilograms of material, and the same applies to the amount of waste produced (kilograms of waste per year). |
| Representativeness | Average data representativeness in line with the defined technological and geographical boundaries. The selected datasets were chosen to reflect the technological and geographical boundaries as accurately as possible. |
| Sources of Data | Administrative data, utility bills, purchase invoices (company source), company software. |

DRAFT

# 3. LIFE CYCLE INVENTORY (LCI)

## 3.1. DATA COLLECTION

The inventory analysis concerns the definition, quantification, and compilation of material and energy flows associated with the production of the studied products.

This phase was conducted in compliance with UNI EN ISO 14044:2021 guidelines. Dedicated editable files were created for data collection purposes. The questionnaires were subsequently sent to Kimfor and completed by the responsible technical staff. The data inventory is attached to this report. Annex A provides the quality assessment of the datasets used (cf. Appendix E, Table E1 of UNI EN 15804:2019). A dataset/LCI dataset is considered "specific" when its quality rating ranges from "very good" to "sufficient."

## 3.2. CALCULATION PROCEDURES

Life Cycle Assessment (LCA) calculations require two main types of information:

1. **Primary data**, referring to the environmental aspects of the system under study, namely the input flows (raw materials, energy, water) and output flows (emissions, waste, products) of the production system. These data are generally provided directly by the company responsible for the study or by suppliers along the value chain.
2. **Impact data**, associated with material and energy flows, necessary for quantifying environmental impacts through impact assessment models. These data are sourced from Life Cycle Inventory (LCI) and Life Cycle Impact Assessment (LCIA) databases, such as Ecoinvent, GaBi, or other internationally recognized databases.

To ensure the reliability and representativeness of the study, the data must meet quality criteria in terms of temporal, geographical, and technological coverage. In particular, data can be classified into the following categories:

- Specific (primary) data

  These are data collected directly from actual processes specific to the system under study. They are typically obtained from:
  - the production site where the studied product is manufactured;
  - other phases of product life cycle, as raw material production or electricity generation. These data may include, for example, information on transportation (distances, means of transport, load factor, fuel consumption), ecc;
  - transport and energy databases (Life Cycle Inventory Data), which are combined with actual data on transport and on the energy parameters of the product.
- Secondary data

  Secondary data originate from external sources and are not specific to the studied system. They represent industrial averages, literature values, or datasets from LCI databases (e.g., Ecoinvent). Examples include:
  - generic data on raw material extraction and processing;
  - national or continental average energy profiles;
  - standardized production processes referring to different geographical or temporal contexts.
- Proxy data

DRAFT

Proxy data constitute a subcategory of secondary data and are used when specific or high-quality secondary data are unavailable. These data are based on estimates, analogies with similar processes, or conservative assumptions. They are applied in a precautionary manner, aiming to avoid underestimating the associated environmental impacts.

## 3.3. REFINEMENT OF SYSTEM BOUNDARIES

The system boundaries and the related information modules investigated were not further refined following previous operations. However, they may be revised in future updates and further development of the study, in line with environmental sustainability objectives and a continuous improvement approach.

## 3.4. ALLOCATION OF INPUT FLOWS AND OUTPUT EMISSIONS

For the modeling of certain primary and secondary data - specifically data derived from activity data processed using appropriate mathematical considerations - the following allocation criterion was applied:

- Physical allocation, based on mass balance relationships or proportions (e.g., the proportion of raw materials constituting the product per specified quantity).

DRAFT

# 4. LIFE CYCLE IMPACT ASSESSMENT (LCIA)

The objective of this phase is to convert the results of the inventory analysis, that is, all input and output flows related to the investigated unit processes, into environmental impact data (i.e., the product's eco-profile) in order to produce reliable and easily understandable results, as well as to assess their magnitude across the investigated information modules. Based on the resulting environmental performance, it is possible to outline the environmental profile of the product under study.

## 4.1. SELECTION OF IMPACT CATEGORIES, CATEGORY INDICATORS, AND CHARACTERIZATION MODELS

For the Life Cycle Impact Assessment (LCIA), the impact categories were selected according to UNI EN 15804:2019 (see section 6.5 of the standard), the reference PCR, and the respective characterization factors defined in UNI EN 15804:2019. The investigated categories are shown in Table 4.

*Table 4: List of impact categories used in accordance with the reference standard and their respective characterization factors*

| IMPACT CATEGORY | CHARACTERIZATION FACTOR ACRONYMS | UNIT |
|---|---|---|
| Climate change | GWP - Total | kg $CO_2$ eq |
| Climate change - Fossil | GWP-fossil | kg $CO_2$ eq |
| Climate change - Biogenic | GWP-biogenic | kg $CO_2$ eq |
| Climate change - Land use and LU change | GWP-luluc | kg $CO_2$ eq |
| Ozone depletion | ODP | kg CFC11 eq |
| Acidification | AP | mol H+ eq |
| Eutrophication, freshwater | EP-freshwater | kg P eq |
| Eutrophication, marine | EP-marine | kg N eq |
| Eutrophication, terrestrial | EP-terrestrial | mol N eq |
| Photochemical ozone formation | POCP | kg NMVOC eq |
| Resource use, minerals and metals | ADP-minerals&metals | kg Sb eq |
| Resource use, fossils | ADP-fossil | MJ |
| Water use | WDP | $m^3$ depriv. |
| Particulate matter | PM | disease inc. |
| Ionising radiation | IRP | kBq U-235 eq |
| Ecotoxicity, freshwater | ETP-fw | CTUe |
| Human toxicity, non-cancer | http-c | CTUh |
| Human toxicity, cancer | http-nc | CTUh |
| Land use | SQP | Pt |
| Global warming (GWP100a) | GWP - GHG | kg $CO_2$ eq |

For the inventory processing and eco-profile calculations, the software SimaPro 10.3.0.1 was used, and the "ECOINVENT 3.10" databases were selected. For the characterization of inventory data with reference to the various impact categories relevant to the system under study, the "EN 15804 +A2 Method" was applied as the calculation method.

## 4.2. CLASSIFICATION

Following the selection of impact categories and calculation models, the various material or energy flows contained in the inventory are allocated to the selected impact categories based on the capacity of the substances or activities to contribute to different environmental issues.

To construct the impact categories, the following damage categories are used:

- Ecology: effects on living species and ecosystems;
- Human health and safety: effects on human health and safety;
- Resources: depletion of energy and material sources;
- Social aspects and habitat degradation: social implications and habitat degradation.

## 4.3. CHARACTERIZATION

Characterization is a step that prepares the calculation of environmental impact.

Impact categories, in a sense, represent the environmental issues associated with the product system. Their selection, therefore, in addition to meeting the goal and scope of the study, must be representative of the physical characteristics of the analyzed product system.

Characterization models, on the other hand, are calculation models developed by dedicated research centers or certified organizations (such as the IPCC) that allow for the development of numerical conversion factors, which are specific and characteristic for each impact category.

During the characterization phase, the method assigns each input or output flow defined in the inventory a factor that allows, for example, the calculation of the relative impact caused by substances emitted into the air and their effects on ecosystems and human health. In this phase, the values of the different impact categories must be expressed in the units specific to each category. The inventory is then processed, and the data are converted into environmental impacts for each pre-selected impact category using a clearly defined unit of measurement, depending on the category itself and enabling comparison. The aggregation of all impacts generated by each input or output element of the inventory, grouped within the same impact category, yields the total impact of that category.

The mathematical relationship used to determine the impacts for each category, with reference to the declared unit, is as follows:

$$Impact_i \left[\frac{kgXeq}{kgproduct}\right] = \sum_{j=1}^{n} \left( FC_i \left[\frac{kgXeq}{kgY}\right] * Inventarydata_j \left[\frac{kgY}{kgproduct}\right] \right)$$

where:

- $Impact_i \left[\frac{kgXeq}{kgproduct}\right]$ is the impact of the i-th impact category measured with reference to the declared unit, where X depends on the category itself;
- $FC_i \left[\frac{kgXeq}{kgY}\right]$ is the characterization factor of the i-th impact category, where Y is the unit of measurement used to measure the inventory data classified in the i-th category;

- $Inventarydata_j \left[\frac{kgY}{kgproduct}\right]$ is the j-th inventory data item that is classified in the i-th impact category, with reference to the declared unit.

DRAFT

# 5. LCA RESULTS

The estimated impact results are only relative statements; they do not indicate endpoints of impact categories, exceedance of threshold values, safety margins, and/or risks.

## 5.1. PRESENTATION OF RESULTS

### 5.1.1. THERMOLOCK THERMAL INSULATION PLASTER

*Table 5: Breakdown of impact assessment results by environmental performance indicators, with reference to the declared unit across the investigated information modules*

| IMPACT CATEGORY | UNIT | A1 -A3 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|
| Global Warming Potential - total (GWP-total) | kg $CO_2$ eq. | 4,50E-01 | 0,00E+00 | 1,65E-02 | 0,00E+00 | 1,99E-02 | 0,00E+00 |
| Global Warming Potential - fossil fuels (GWP-fossil) | kg $CO_2$ eq. | 4,08E-01 | 0,00E+00 | 1,64E-02 | 0,00E+00 | 1,96E-02 | 0,00E+00 |
| Global Warming Potential - biogenic (GWP-biogenic) | kg $CO_2$ eq. | 4,18E-02 | 0,00E+00 | 8,88E-05 | 0,00E+00 | 2,31E-04 | 0,00E+00 |
| Global Warming Potential - land use and land use change (GWP-luluc) | kg $CO_2$ eq. | 2,87E-04 | 0,00E+00 | 6,49E-06 | 0,00E+00 | 4,88E-05 | 0,00E+00 |
| Depletion potential of the stratospheric ozone layer (ODP) | kg CFC-11 eq. | 4,88E-09 | 0,00E+00 | 2,28E-10 | 0,00E+00 | 3,82E-10 | 0,00E+00 |
| Acidifcation potential, Accumulated Exceedance (AP) | mol H+ eq. | 1,97E-03 | 0,00E+00 | 5,45E-05 | 0,00E+00 | 1,07E-04 | 0,00E+00 |
| Eutrophication potential - freshwater (EP-freshwater) | kg P eq. | 8,08E-05 | 0,00E+00 | 1,28E-06 | 0,00E+00 | 1,03E-04 | 0,00E+00 |
| Eutrophication potential - marine (EP-marine) | kg N eq. | 5,01E-04 | 0,00E+00 | 1,77E-05 | 0,00E+00 | 4,07E-05 | 0,00E+00 |
| Eutrophication potential - terrestrial (EP-terrestrial) | mol N eq. | 5,48E-03 | 0,00E+00 | 1,92E-04 | 0,00E+00 | 4,41E-04 | 0,00E+00 |
| Photochemical Ozone Creation Potential (POCP) | kg NMVOC eq. | 1,89E-03 | 0,00E+00 | 7,59E-05 | 0,00E+00 | 1,51E-04 | 0,00E+00 |
| Abiotic depletion potential - non-fossil resources (ADPE)** | kg Sb eq. | 5,14E+00 | 0,00E+00 | 2,29E-01 | 0,00E+00 | 3,31E-01 | 0,00E+00 |
| Abiotic depletion potential - fossil resources (ADPF)** | MJ, net calorific value | 3,90E-06 | 0,00E+00 | 5,24E-08 | 0,00E+00 | 5,60E-08 | 0,00E+00 |
| Water (user) deprivation potential (WDP)** | $m^3$ world eq. deprived | 8,41E-02 | 0,00E+00 | 1,04E-03 | 0,00E+00 | 1,04E-02 | 0,00E+00 |
| Particulate Matter emissions (PM) | Disease incidence | 2,31E-08 | 0,00E+00 | 1,30E-09 | 0,00E+00 | 2,35E-09 | 0,00E+00 |
| Ionizing radiation, human health (IRP)* | kBq U235 eq. | 1,07E-02 | 0,00E+00 | 1,88E-04 | 0,00E+00 | 2,96E-04 | 0,00E+00 |
| Eco-toxicity - freshwater (ETP-fw)** | CTUe | 2,78E+00 | 0,00E+00 | 6,11E-02 | 0,00E+00 | 2,25E-01 | 0,00E+00 |
| Human toxicity, cancer effect (HTP-c)** | CTUh | 1,59E-09 | 0,00E+00 | 8,47E-11 | 0,00E+00 | 1,53E-10 | 0,00E+00 |
| Human toxicity, non-cancer effects (HTP-nc)** | CTUh | 6,15E-09 | 0,00E+00 | 1,43E-10 | 0,00E+00 | 4,41E-10 | 0,00E+00 |
| Land use related impacts/Soil quality (SQP)** | Pt | 1,59E-09 | 0,00E+00 | 1,37E-01 | 0,00E+00 | 8,10E-01 | 0,00E+00 |

*The results of this environmental impact indicator should be used with caution, as the uncertainties associated with these results are high or due to limited experience with this indicator (cf. UNI EN 15804:2019).

** This impact category mainly addresses the potential impact of low-dose ionizing radiation on humans from the nuclear fuel cycle. It does not consider effects due to potential nuclear accidents, occupational exposure, or the disposal of nuclear waste in underground facilities. Potential ionizing radiation from the soil, radon, and some building materials is also not measured by this indicator.

*Table 6: The indicator includes all greenhouse gases covered by total GWP but excludes the uptake and emissions of biogenic carbon dioxide and the biogenic carbon stored in the product. Therefore this indicator is equivalent to the GWP indicator originally defined in EN 15804:2012 + A1:2013.*

DRAFT

| Potential environmental impacts – additional indicator | UNIT | A1-A3 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|
| GWP - GHG | Kg $CO_2$ eq | 4,06E-01 | 0,00E+00 | 1,63E-02 | 0,00E+00 | 1,95E-02 | 0,00E+00 |

*Table 7: Breakdown of resource use results with reference to the declared unit across the investigated information modules*

| PARAMETERS | UNIT | A1-A3 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|
| PERE | MJ | 1,00E+00 | 0,00E+00 | 2,90E-03 | 0,00E+00 | 4,83E-03 | 0,00E+00 |
| PERM | MJ | 5,52E-01 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| PERT | MJ | 1,56E+00 | 0,00E+00 | 2,90E-03 | 0,00E+00 | 4,83E-03 | 0,00E+00 |
| PENRE | MJ | -1,09E-01 | 0,00E+00 | 2,77E-04 | 0,00E+00 | 5,07E-03 | 0,00E+00 |
| PENRM | MJ | 1,26E-01 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| PENRT | MJ | 1,65E-02 | 0,00E+00 | 2,77E-04 | 0,00E+00 | 5,07E-03 | 0,00E+00 |
| SM | Kg | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| RSF | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| NRSF | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| FW | $m^3$ | 2,25E-03 | 0,00E+00 | 3,06E-05 | 0,00E+00 | 2,53E-04 | 0,00E+00 |

*Table 8: Breakdown of waste flows with reference to the declared unit across the investigated information modules*

| PARAMETERS | UNIT | A1-A3 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|
| HWD | kg | 5,19E-05 | 0,00E+00 | 1,58E-06 | 0,00E+00 | 7,64E-06 | 0,00E+00 |
| NHWD | kg | 7,62E-02 | 0,00E+00 | 1,08E-02 | 0,00E+00 | 1,22E-02 | 0,00E+00 |
| RWD | kg | 3,03E-06 | 0,00E+00 | 4,62E-08 | 0,00E+00 | 7,26E-08 | 0,00E+00 |

*Table 9: Distribution of output flows with reference to the declared unit across the investigated information modules*

| PARAMETERS | UNIT | A1-A3 | C1 | C2 | C3 | C4 | D |
|---|---|---|---|---|---|---|---|
| CRU | Kg | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| MFR | Kg | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| MER | kg | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| EEE | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |
| EET | MJ | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 | 0,00E+00 |

DRAFT

# 6. BIOGENIC CARBON CONTENT IN THE PRODUCT AND ITS PACKAGING

*Table 10: Biogenic carbon content in the Thermolock thermal insulation plaster product and its packaging*

| | C BIOGENIC | |
|---|---|---|
| | Product | Packaging |
| Thermolock Thermal Insulation Plaster | not relevant | 1,54E-02 |

The formula for calculating biogenic carbon is based on the EN 16449:2014 standard:

$$P_{CO2} = \frac{44}{12} x c_f x \frac{\rho_\omega x V_\omega}{1+\frac{\omega}{100}}$$

where:

- $C_f$: is the carbon fraction in the packaging (set at 16.28%);
- ω is the moisture content in the packaging (set at 9.30%);
- $P_\omega$ is the density of the packaging;
- $V_\omega$ is the volume of the packaging.

For the above biogenic carbon calculation, the conversion factor 44/12 was not applied, thus obtaining the biogenic carbon itself and not its conversion to $CO_2$.

DRAFT

# 7. RESULTS ANALYSIS AND AVERAGE PERCENTAGE CONTRIBUTIONS

The work presented here highlights how it is possible to analyse the products under study along the information modules investigated (A1 – A3, C1-C2-C3-C4 + D). The approach adopted by the company and the dynamic calculation model will allow Kimfor to have a system capable of monitoring how a change in an input/output parameter of a unit process can vary its contribution in terms of environmental impact and, therefore, to identify those significant and controllable parameters that can lead to a reduction in the resulting environmental risks. The following analysis shows that the greatest impact came from modules A1-A3, linked to upstream production processes in the supply chain, with a respective overall impact percentage of 78%, with the largest contribution coming from DLP 2140.

*Table 11: Contributions Analysis*

| A1-A3 | C1 | C2 | C3 | C4 |
|---|---|---|---|---|
| 78% | 0% | 2% | 0% | 11% |

*Figure 5: Average incidence percentage, average incidence percentage for each module*

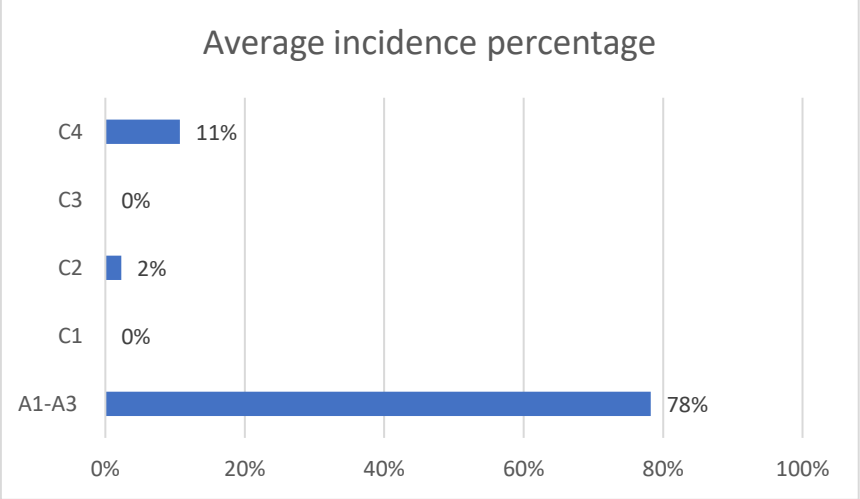
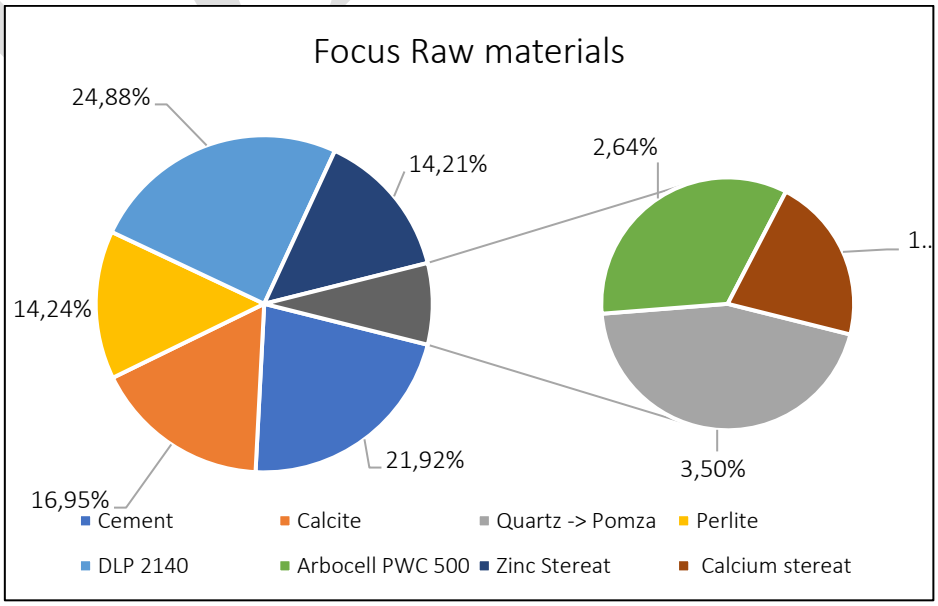



*Table 12: Focus on Modules A1-A3*

| | Cement | Calcite | Quartz -> Pomza | Perlite | DLP 2140 | Arbocell PWC 500 | Zinc Stereat | Calcium stereat |
|---|---|---|---|---|---|---|---|---|
| Climate change | 42,22% | 13,57% | 2,41% | 8,91% | 23,72% | 1,20% | 5,04% | 2,93% |
| Climate change - Fos-sil | 42,32% | 13,61% | 2,19% | 8,95% | 23,87% | 1,08% | 5,01% | 2,97% |
| Climate change - Bio-genic | 37,52% | 11,30% | 13,71% | 7,06% | 16,29% | 6,84% | 6,17% | 1,10% |
| Climate change - Land use and LU change | 19,07% | 20,87% | 3,17% | 16,99% | 25,00% | 2,58% | 9,92% | 2,39% |
| Ozone deple-tion | 12,31% | 15,33% | 1,58% | 9,61% | 51,83% | 1,04% | 6,47% | 1,82% |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Acidification | 24,71% | 17,11% | 3,21% | 23,74% | 16,95% | 1,66% | 11,20% | 1,42% |
| Eutrophication, freshwater | 26,45% | 18,56% | 2,62% | 7,21% | 29,58% | 1,93% | 11,53% | 2,12% |
| Eutrophication, marine | 29,52% | 14,85% | 2,97% | 30,65% | 14,65% | 1,52% | 4,54% | 1,29% |
| Eutrophication, terrestrial | 29,66% | 14,71% | 2,89% | 31,99% | 13,69% | 1,42% | 4,40% | 1,24% |
| Photochemical ozone formation | 24,65% | 16,42% | 2,73% | 25,99% | 22,59% | 1,37% | 4,84% | 1,41% |
| Resource use, fossils | 17,17% | 14,87% | 2,01% | 9,46% | 47,62% | 1,12% | 5,95% | 1,78% |
| Resource use, minerals and metals | 9,30% | 27,88% | 0,41% | 1,77% | 14,09% | 17,64% | 27,76% | 1,15% |
| Water use | 10,79% | 29,80% | 11,89% | 3,62% | 29,08% | 1,06% | 12,61% | 1,14% |
| Particulate matter | 22,80% | 16,46% | 6,09% | 22,74% | 20,01% | 2,28% | 8,37% | 1,26% |
| Ionising radiation | 15,59% | 20,52% | 1,20% | 8,32% | 34,88% | 2,37% | 13,75% | 3,37% |
| Ecotoxicity, freshwater - part 1 | 18,41% | 18,44% | 3,13% | 22,04% | 26,12% | 1,96% | 8,38% | 1,51% |
| Ecotoxicity, freshwater - part 2 | 6,36% | 5,12% | 0,32% | 2,11% | 26,26% | 0,41% | 59,07% | 0,35% |
| Human toxicity, non-cancer | 14,73% | 12,08% | 0,93% | 3,12% | 13,64% | 1,10% | 53,55% | 0,85% |
| Human toxicity, cancer | 12,89% | 20,49% | 3,04% | 26,26% | 22,81% | 1,63% | 11,42% | 1,46% |
| Land use | 35,40% | 30,37% | 27,96% | -44,46% | 34,72% | 2,52% | 10,98% | 2,50% |
| Media materie | 21,92% | 16,95% | 3,50% | 14,24% | 24,88% | 2,64% | 14,21% | 1,66% |

*Graph 1: Focus on raw materials*




DRAFT

# 8. CONCLUSIONS, LIMITATIONS, AND RECOMMENDATIONS

The work presented here demonstrates the feasibility of analyzing the studied products across the investigated information modules [A1-A3, C1-C4 + D]. The approach adopted by the company and the dynamic calculation model will enable Kimfor to have a system capable of monitoring how changes in input/output parameters of a unit process affect its contribution in terms of environmental impact. This, in turn, allows for the identification of significant and controllable parameters that can lead to a reduction in the resulting environmental risks.

In accordance with the methodological approach defined by ISO 14040 and ISO 14044 standards, and consistent with the "Construction Products" PCR version 2.0.1, the analysis of results shows that the predominant contribution to environmental impacts throughout the life cycle of the analyzed products is attributable to the material component (within modules A1-A3), with a particularly significant contribution from DLP 2140 (24,88%). This confirms the strategic importance of material selection during the design phase, with a recommendation to use materials containing pre- and post-consumer recycled content, in line with circular economy principles. The adoption of such criteria can substantially contribute to the overall reduction of environmental impacts and the improvement of environmental performance.

In Kimfor's interactive context with its stakeholders, both collective and individual, communicating its environmental management policy regarding product manufacturing becomes crucial and synonymous with a company that aims to be perceived as reliable and transparent. The study described in this report enabled the company to identify the emission hotspots characterizing the production and processing of the analyzed product. This provides a solid foundation for implementing targeted and concrete action plans aimed at reducing and containing environmental impacts.

It should be noted that the calculation model is scalable and can be updated, including the integration of potential new relevant unit processes.

Finally, the company intends to continue integrating this study into future design phases, in order to associate specific products with a dedicated monitoring system, supporting the environmental sustainability of its operations and continuous improvement.

DRAFT

## BIBLIOGRAPHY

ISO (UNI EN ISO 14040). Environmental management, Life Cycle Assessment – Principles and Framework. Ginevra: International Organization for Standardization (ISO)

ISO. (UNI EN ISO 14044). Environmental management, Life Cycle Assessment – Requirements and Guidelines. Ginevra: International Organization for Standardization (ISO)

EN 15804:2012+A2:2019, Sustainability of construction works - Environmental product declarations - Core rules for the product category of construction products.

General Programme Instructions for the International EPD® System. Version 5.0.1

PCR Construction Products, PCR 2019:14, v2.0.1

## WEB REFERENCES

https://cop23.com.fj/

www.carbonfootprint.com

http://ec.europa.eu

www.enea.it

www.epa.gov

www.europarl.europa.eu

www.environdec.com/it/

http://www.pnas.org/

https://thermolock.com/en/products/kimfor

DRAFT

# ANNEX A: DATA AND DATASET QUALITY

The following section reports the quality assessment of the datasets used (see Appendix E, Table E1 of UNI EN 15804:2019). From "very good" to "sufficient," an LCI dataset/data is considered specific.

*Table 13: Criteria used to define the data quality*

| QUALITY LEVEL | GEOGRAPHICAL REPRESENTATIVENESS | TECHNICAL REPRESENTATIVENESS | TEMPORAL REPRESENTATIVENESS |
|---|---|---|---|
| Very good | Data from the area under study | Data from processes and products under study. Same state of the art as specified in the objective and scope and field of application (i.e. identical technology). | Less than 3 years difference between the reference year according to the documentation and the period of time for which the data are representative |
| Good | Average data from a larger area that includes the area under study | Data from processes and products under study (with similar technology). Evidence of deviations in the state of the art, e.g. different by-products. | Less than 6 years difference between the reference year according to the documentation and the period of time for which the data are representative |
| Fair | Data from areas with similar production conditions | Data from processes and products under study but from different technologies. This score is also applied when the technology is not specified; for example, wheat (no further specification) | Less than 10 years difference between the reference year according to the documentation and the period of time for which the data are representative |
| Poor | Data from areas with slightly different production conditions | Data on related processes or products; organic wheat under study, data for organic rye provided. | Less than 15 years difference between the reference year according to the documentation and the period of time for which the data are representative |
| Very poor | Data from unknown or significantly different areas (North America instead of the Middle East, OECD-Europe instead of Russia) | Data on related processes but on a different scale or from a different technology; organic wheat under study, data for conventional wheat provided. | Age of data unknown or greater than 15 years difference between the reference year according to the documentation and the period of time for which the data are representative |

The following scale of values is then provided in order to verify the level of each dataset individually.

| Process name | Source type | Source | Relevance year | Data category | Share of data, of GWP-GHG results for A1-A3 | Relevance | Temporal representativeness | Technical representativeness | Geographical representativeness | DQR | Quality rating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Cement | Company data | Ecoinvent 3.10 | 2024 | secondary | 14% | 14% for GWP GHG; for others from 2% (Land use) to 31% (Climate change - Fossil) | 2 | 2 | 2 | 2 | Very good quality |
| Calcite | Company data | Ecoinvent 3.10 | 2024 | secondary | 11% | 11% for GWP GHG; for others from 2% (Climate change - Biogenic) to 25% (Resource use, minerals and metals) | 2 | 2 | 2 | 2 | Very good quality |

DRAFT

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Perlite | Company data | Ecoin-vent 3.10 | 2024 | secondary | 9% | 9% for GWP GHG; for others from 2% (Land use) to 24% (Eutrophication, ter-restrial) | 2 | 2 | 2 | 2 | Very good quality |
| DLP 2140 | Company data | Ecoin-vent 3.10 | 2024 | secondary | 16% | 16% for GWP GHG; for others from 2% (Land use) to 36% (Ozone depletion) | 2 | 2 | 2 | 2 | Very good quality |
| Zinc Stereat | Company data | Ecoin-vent 3.10 | 2024 | secondary | 10% | 10% for GWP GHG; for others from 0,56% (Land use) to 53% (Ecotoxicity, freshwater - part 2) | 2 | 2 | 2 | 2 | Very good quality |
| Packaging | Company data | Ecoin-vent 3.10 | 2024 | secondary | 8% | 8% for GWP GHG; for others from 0,86% (Resource use, minerals and metals) to 58% (Climate change - Biogenic) | 2 | 2 | 2 | 2 | Very good quality |
| Upstream raw material tran-sport | Company data | Ecoin-vent 3.10 | 2024 | primary | 12% | 12% for GWP GHG; for others from 1% (Climate change - Bi-ogenic) to 22% (Par-ticulate matter) | 2 | 2 | 2 | 2 | Very good quality |
| | | | | | | | | | | 2,08 | Good quality |

*Table 14: Attribution scale regarding data quality*

| QUALITY | FINAL SCORE |
|---|---|
| Very good | ≤1,6 |
| Good | >1,6 to 2 |
| Fair | 2 to 3 |
| Poor | 3 to 4 |
| Very poor | ≥4 |

DRAFT